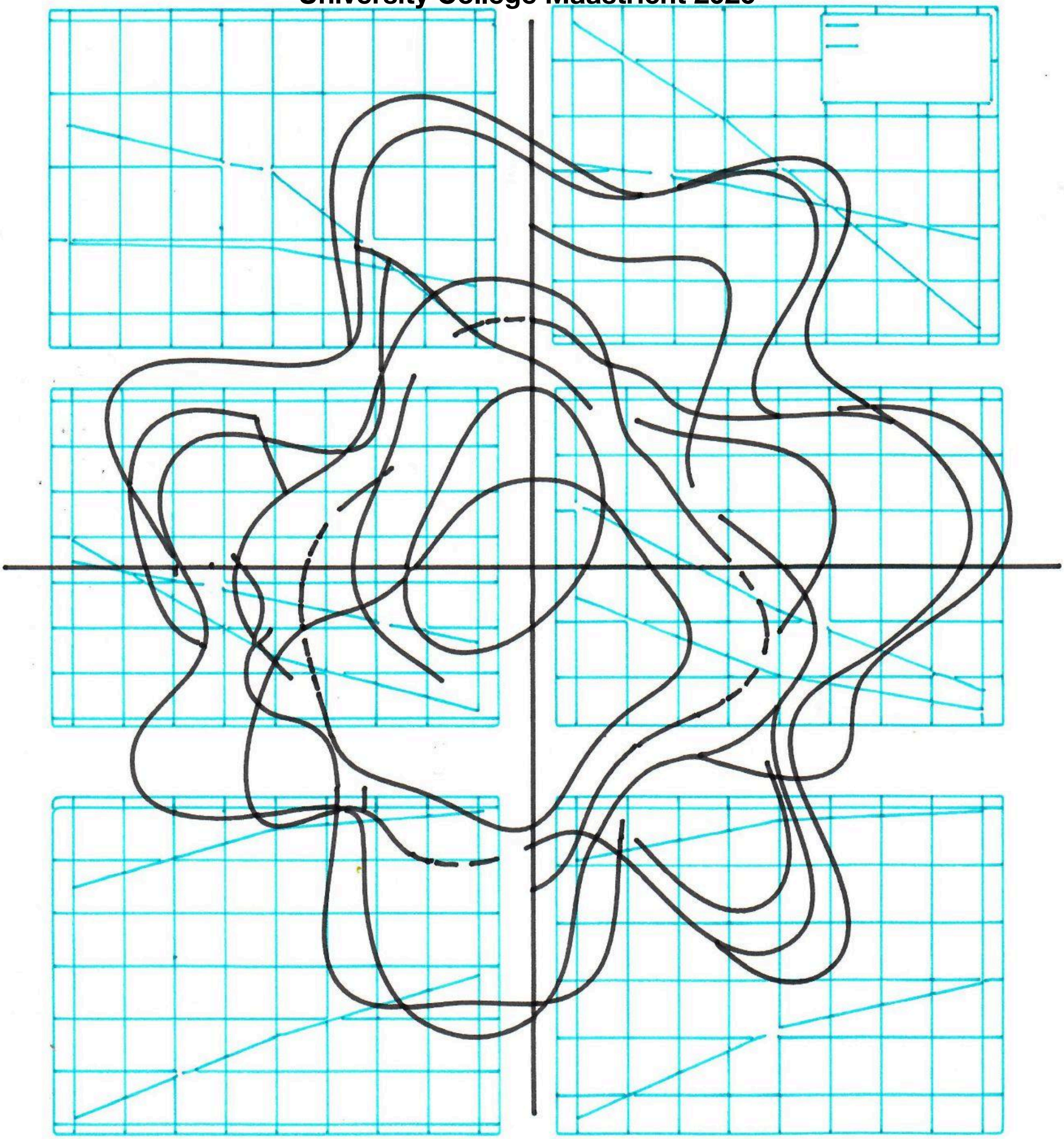
Capstone

Training for Robustness: A Cost-Benefit Analysis of Degradation Augmentation in CNN Image Classification

University College Maastricht 2026



Abstract

Convolutional neural networks lose accuracy when their inputs differ from clean training data. This is a significant problem for deployed systems. This capstone compares three training-time data augmentation strategies for improving CNN robustness to image degradation at CIFAR-10 scale. Two architectures (ResNet-18 and MobileNet-V2) are trained on CIFAR-10 across five seeds and evaluated on six degradation types at three severity levels each: Gaussian noise, Gaussian blur, motion blur, JPEG compression, contrast reduction and darkening. Three strategies are compared: single-type augmentation on one degradation, mixed augmentation across the full 18-condition grid and severity-curriculum augmentation with a linear ramp from mild to severe over 100 epochs. Statistical comparisons use paired t-tests with Bonferroni correction.

Mixed augmentation produces the highest mean robustness on both architectures, gaining approximately 23 pp of robustness for 3 pp of clean accuracy lost. Single-type training improves its target degradation but introduces negative transfer to other conditions like contrast and darkening. Severity-curriculum augmentation does not produce the gains predicted by the curriculum learning hypothesis at this scale. With Gaussian blur it does the opposite: a near-total collapse to near-random performance on the clean test set (12.77% on ResNet-18, 11.86% on MobileNet-V2). The mechanism is identified as catastrophic forgetting interacting with a degradation which destroys spatial structure at a 32×32 resolution. The practical takeaway is that mixed augmentation is the most efficient strategy in the clean-robustness tradeoff and that naive curriculum schedules can fail catastrophically.

1. Introduction

1.1 Background and Motivation

Computer vision systems are increasingly being deployed in environments that look nothing like the controlled conditions their models were trained on. Self-driving cars, medical imaging pipelines, surveillance cameras, satellite imagery analysis. They all meet messy real-world inputs: sensor noise, blur, lossy compression, low contrast, uneven illumination and more. Classifiers trained on clean data tend to lose accuracy when faced with these conditions and the drop can be substantial like Hendryck's and Dietterich (2019) have shown. Robustness to such degradations is not optional. It is a deployment requirement.

This capstone builds directly on a previous benchmarking experiment in which two widely used lightweight CNN architectures, ResNet-18 (He et al., 2016) and MobileNet-V2 (Sandler et al., 2018) were trained on CIFAR-10 and evaluated under controlled image degradations. The earlier work established a baseline against which augmentation strategies can be measured.

An early run was suggesting that MobileNet-V2 was actually beating ResNet-18 on JPEG compression. A more rigorous replication with 200 epochs and five seeds showed that this finding was a result of undertraining rather than a real architectural advantage. To summarise: robustness comparisons made before convergence can flip the architecture ranking entirely. With proper training the picture becomes clearer, but more concerning. ResNet-18 reaches 95.09% ± 0.17% clean accuracy but drops to 25.03% under Gaussian blur and 61.96% under Gaussian noise, both averaged across severities. MobileNet-V2 follows the same pattern with sharper drops. The gap between clean and degraded performance is considerably large and unsolved. This capstone is trying to close it by systematically comparing three augmentation strategies.

1.2 Research Gap

The literature on data augmentation is huge (Shorten & Khoshgoftaar, 2019), and corruption robustness benchmarks are well established (Hendrycks & Dietterich, 2019). Methods such as AugMix (Hendrycks et al., 2020) and RandAugment (Cubuk et al., 2020) have shown that training time augmentation improves robustness. At the top of the field the picture is well established, but three gaps remain.

First, most existing work is at ImageNet scale with larger models. The combination of CIFAR-scale data and the imaging-engineering style degradations used here is sparsely covered. Second, there are few systematic comparisons of simpler degradation-specific strategies under matched conditions. Third, the cost side is underreported. Robustness gains are often quoted without the corresponding clean accuracy penalty, which is precisely the tradeoff that matters for deployment.

A related issue is that AugMix and RandAugment combine several mechanisms: augmentation chaining, consistency regularisation and learned magnitudes all get folded into the same method, so it is hard to tell which component contributes what exactly.

1.3 Research Questions

Primary question. Does training with the same degradation used at evaluation improve CNN robustness to those degradations? What does it cost in clean image accuracy?

Sub-questions. (SQ1) which augmentation strategy (single-type, mixed, or severity-curriculum) achieve the highest robustness gain per unit of clean accuracy lost? (SQ2) Do ResNet-18 and MobileNet-V2 respond differently to the same augmentation strategy? (SQ3) For the two degradations which cause the largest baseline accuracy drops (Gaussian blur and Gaussian noise), does single-type augmentation transfer robustness improvement to other degradation types?

1.4 Hypotheses

The study tests four hypotheses, each tied to one of the sub-questions above. Each hypothesis is stated as a null hypothesis (H0) and an alternative hypothesis (HA). All paired tests use a Bonferroni corrected significance threshold of $\alpha = 0.05/7 \approx 0.007$. A significant result in the predicted direction rejects H0 in favour of HA. A non-significant result means H0 is not rejected.

H1 (SQ1). H0: mixed augmentation does not produce higher mean robustness than single-type and severity-curriculum augmentation. HA: mixed augmentation produces higher mean robustness than both other strategies on both architectures.

H2 (SQ1). H0: severity-curriculum augmentation, expressed as a linear severity ramp on a single degradation type at 32×32 resolution, does not produce higher accuracy on the trained degradation than single-type augmentation paired by seed. HA: curriculum augmentation produces higher accuracy on the trained degradation than single-type augmentation.

H3 (SQ2). H0: ResNet-18 and MobileNet-V2 do not differ in their absolute gain from mixed augmentation. HA: ResNet-18 gains more in absolute terms than MobileNet-V2. This is the formal hypothesis under test. Further qualitative observations are reported in Section 5.

H4 (SQ3). H0: single-type training does not produce both a statistically significant improvement on the trained degradation and a statistically significant decrease on another degradation type. HA: single-type training produces a statistically significant improvement on the trained degradation and a statistically significant decrease on at least one other degradation type. Both directions must hold for HA to be supported.

1.5 Contributions

This capstone contributes a robustness benchmark validated across five seeds for ResNet-18 and MobileNet-V2 on CIFAR-10, evaluated on six degradation types at three severity levels each, with paired t-tests and Bonferroni correction. Comparisons between architectures therefore rest on statistically defensible evidence rather than single-run variation. A second contribution is a systematic comparison of three augmentation strategies, single-type, mixed and severity-curriculum, under that same framework. Identical training setup, identical seeds, identical evaluation grid. Strategy-level differences therefore reflect the strategy itself rather than uncontrolled variation in the surrounding pipeline.

2. Related Work

2.1 CNN Architectures

ResNet-18 (He et al., 2016) introduced residual connections, which allowed stable training of very deep networks. The skip connections let gradients flow back through the layers, training networks beyond thirty layers becomes practical. MobileNet-V2 (Sandler et al., 2018) follows a different design path. It uses depthwise separable convolutions and inverted residual blocks. The result is way fewer parameters and a model designed for compute-constrained, edge-side deployment.

Both architectures are well studied on clean benchmarks. Their behaviour under distributional shift at full convergence is less explored. That is the gap this capstone addresses.

2.2 Robustness Benchmarking

Hendrycks and Dietterich (2019) introduced ImageNet-C; a benchmark covering 15 corruption types at 5 severity levels. Their central finding was that SOTA clean accuracy does not imply robustness to corruption. The two metrics can diverge sharply. The methodological framing they use, evaluating clean and corrupted performance separately is the foundation for the benchmark used in this study. Geirhos et al. (2019) added a complementary result. ImageNet-trained CNNs show a strong texture bias, which helps explain why corruptions that disrupt texture cause disproportionate accuracy drops. The texture-bias argument comes back in Section 5.2

2.3 Data Augmentation

Standard geometric augmentation such as random cropping and horizontal flipping help generalisation but doesn't target the distributional shift that degradations introduce. Domain-specific augmentation, that is, training on corrupted versions of the input, narrows the gap between clean and degraded performance directly. AugMix (Hendrycks et al., 2020) chains augmentation operations and adds a consistent regularisation term, and performs strongly on ImageNet-C. RandAugment (Cubuk et al., 2020) takes a simpler route with a single uniform magnitude parameter and no search procedure. Both methods confirm two points. (1) Augmentation strategy design matters. (2) interference between augmentation types is a real phenomenon which has to be controlled for

2.4 Curriculum Learning

Bengio et al. (2009) formalised the curriculum learning hypothesis, which states that training on examples ordered from easy to hard improves both optimisation and generalisation compared to random ordering. The intuition is that the model first learns the basic structure of the problem, then handles harder cases once a useful representation is in place. Applied to degradation augmentation, this suggests starting with mild corruption and ramping up to severe over the course of training.

The schedule used here is a linear severity ramp from level 0 to max over epochs 1 to 100, held at max for epochs 100 to 200, applied to one degradation type at a time. This schedule was selected because it is parameter free and intuitive. Other schedules exist: cosine ramps, stepped schedules and validation-gated approaches which adjust difficulty based on training progress. Whether the choice of schedule matters at CIFAR-10 scale is one of the questions this study addresses; the result, examined in Section 5.2, is that it does.

2.5 Statistical Rigour

Bouthillier et al. (2021) showed that variance across random seeds in machine learning benchmarks is large enough that single-run comparisons cannot be trusted. The benchmark this capstone builds on uses 5 seeds, paired t-tests, and Bonferroni correction precisely for this reason. The same framework is reused here for the augmentation strategy comparisons, so that strategy-level claims rest on statistically valid evidence rather than single-run flukes.

3. Methodology

3.1 Research Design

This is quantitative experimental research conducted in the empirical machine learning paradigm. Hypotheses about model behaviour are tested by training and evaluating models under controlled, reproducible conditions. There are no human subjects involved. CIFAR-10 is publicly available and ethically unproblematic. The main ethical consideration is computational cost. The entire experimental matrix consumed approximately 60 to 75 GPU-hours on a single A100, which is reported here in the interest of responsible reporting. The author paid the computer costs privately.

3.2 Dataset

CIFAR-10 contains 60,000 colour images of size 32×32 across 10 classes, with 50,000 training images and 10,000 test images (Krizhevsky, 2009). The classes are airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck. CIFAR-10 was selected for several reasons. It is the standard benchmark for this class of experiment, large enough to train non-trivial models but small enough to allow the multiple-seed runs needed for statistical validity within a 1 semester time frame. CIFAR-100 was considered but not used: the smaller per-class sample size would have increased seed-level variance and reduced the statistical power of the comparisons. TinyImageNet would have permitted closer alignment with prior work but was not feasible within the available compute budget. The full ImageNet-C benchmark used by Hendrycks and Dietterich (2019) was impossible at this resource level.

3.3 Degradation Conditions

Six degradation types are applied at three severity levels each, identical to the baseline benchmark, giving an 18-condition evaluation grid (Table 1). The six degradations cover sensor noise, optical blur, compression artefacts, and lighting shifts. They are a reasonable amount of the failure modes which deployed imaging systems are exposed to. Keeping this grid unchanged from the baseline benchmark guarantees direct comparability with prior results.

Category	Degradation	Level 1	Level 2	Level 3
Noise	Gaussian noise (σ)	0.02	0.05	0.10
Blur	Gaussian blur (kernel)	3	5	7
Blur	Motion blur (kernel)	3	5	7
Compression	JPEG (quality)	50	30	10
Lighting	Contrast (factor)	0.8	0.6	0.4
Lighting	Darkening (factor)	0.8	0.6	0.4

Table 1. Degradation grid. Six degradation types at three severity levels each, giving 18 evaluation conditions.

3.4 Augmentation Strategies

Three strategies are compared, all applied on top of standard geometric augmentation (random crop with padding 4, random horizontal flip). Degradation augmentation is applied after geometric augmentation so the degradations affect the cropped region rather than the original image. All strategies degrade 100% of the training images.

For single-type augmentation, each training image is degraded with a single fixed degradation type, with severity randomly sampled from the corresponding row of the grid. Due to compute constraints, this is tested only for Gaussian blur and Gaussian noise, the two degradations which caused the most severe baseline collapse. The matrix is 2 types × 2 architectures × 5 seeds = 20 runs.

For mixed augmentation, each training image is degraded with a degradation type and severity both sampled uniformly at random from the full grid. This tests whether broad exposure produces broad robustness: 2 architectures × 5 seeds = 10 runs. Uniform sampling across the grid was chosen as the simplest baseline; weighted sampling schemes for example, oversampling milder severities were not tested. This is acknowledged as a methodological choice and revisited in Section 5.6.

For severity-curriculum augmentation: a single degradation type is used with severity ramping linearly from mildest to hardest over the first 100 epochs (out of 200), then held at maximum for the remaining 100. This tests whether ordered exposure outperforms random exposure for the same degradation: 2 types × 2 architectures × 5 seeds = 20 runs. The linear ramp was selected for its simplicity and parameter-free design. Others like cosine, stepped, and validation-gated variants were not tested and are noted as a limitation in Section 5.6.

3.5 Model Training

Both architectures are adapted for 32×32 input. ResNet-18 uses a 3×3 initial convolution with no maxpool layer. MobileNet-V2 uses stride-1 in its initial convolution. Training uses SGD with momentum 0.9, weight decay 5×10^{-4}, initial learning rate 0.1, and a MultiStepLR learning rate schedule which decays by ×0.1 at epochs 100 and 150. all models train for 200 epochs with batch size 128 on a single GPU (Google Colab A100). The training pipeline was implemented in PyTorch with torchvision (Paszke et al., 2019). All experiment code was written for this project.

Two choices here are worth motivating. Label smoothing was not used. The SGD + MultiStepLR + cross-entropy combination is the standard convergence recipe for these architectures on CIFAR-10, and adding label smoothing would have added a second moving piece when comparing strategies. There is also no held-out validation split. This was deliberate and planned. The 200-epoch schedule used the hyperparameters above, with no early stopping, and the final-epoch checkpoint is reported. Using a validation split for model selection would have leaked some training-time information into the test pipeline, and would have shrunk the training set across an already tight compute matrix. The tradeoff is that there is no in-training signal for over- or under-fitting. That cost is flagged again in Section 5.6.

3.6 Experimental Protocol

Five random seeds (0, 42, 123, 456 , 789) control parameter initialisation, training batch ordering, and augmentation sampling. The full experimental matrix comprises 60 training runs: 10 baseline, 10 mixed, 20 single-type, and 20 curriculum. Seeds are matched between baseline and augmented models so that paired statistical tests can be applied.

3.7 Evaluation

Each trained model is evaluated on clean test accuracy and on degraded accuracy across all 18 degradation-severity combinations. The primary summary measure is mean robustness, defined as the average accuracy across all 18 degraded conditions. Secondary measures include per-degradation accuracy (averaged over severity levels) and the robustness-accuracy tradeoff, defined as mean robustness gain divided by clean accuracy cost and analysed in Section 5.4.

3.8 Statistical Analysis

Paired t-tests compare augmented and baseline models and they are paired with the seed which eliminates seed level effects. Each strategy is compared seven times: one is pure precision, and the other six are the average of the severity of the six degradation types. The Bonferroni correction is used to correct the multiple number of comparisons and gives $0.05/7 \approx 0.007$.

Cohen's d is reported in addition to the p-values. One thing to note about the d values: the variance between seeds is very low (they all lie within the 0.07 percent range) thus making the d value unusually large. This is due to the small denominator, not the effect size. The effects are not imaginary but the magnitude should be read as the indicator of precise measurement as opposed to overwhelmingly large effect size (Bouthillier et al., 2021).

3.9 Code and Data Availability

All code used in this study, including training scripts, evaluation pipelines, analysis notebooks and figure generation, is available at https://github.com/Chaospossum/cv-robustness-thesis. CIFAR-10 is publicly available through torchvision. Together with the training details in Sections 3.5 and 3.6, this provides the information needed to replicate the full experimental matrix.

4. Results

4.1 Baseline

Both architectures perform well on clean images but degrade severely under blur and noise, while contrast and darkening produce only mild drops. This section quantifies that pattern.

Clean accuracy is 95.09% ± 0.17% for ResNet-18 and 89.18% ± 0.07% for MobileNet-V2. Both models are fully converged and consistent with reported results for these architectures on CIFAR-10. Performance deteriorates sharply at severity 3. Gaussian blur reduces ResNet-18 to 17.48% and MobileNet-V2 to 15.18%, which is close to the 10% chance level for a ten-class problem. Gaussian noise is not random but is close: ResNet-18 24.97% and MobileNet-V2 17.41%. Both models keep contrast and darkening at severity 3 above 73%, which is mild compared to blur and noise.

The average degraded accuracy (mean across all 18 degraded conditions) is 65.59% ± 0.78% for ResNet-18 and 58.80% ± 1.19% for MobileNet-V2. This is a drop of approximately 30pp from clean accuracy in both cases. The difference is considerably large and is the whole motive behind the augmentation experiments which follow. Full results by state are presented in Table 2. The pattern, with blur and noise as the most damaging conditions and lighting shifts as the mildest, is consistent with Dodge and Karam (2016), who found blur and noise to be the quality reductions which deep networks tolerate worst.

	R18 Sev1	R18 Sev2	R18 Sev3	MNv2 Sev1	MNv2 Sev2	MNv2 Sev3
Clean	95.09 ± 0.17	—	—	89.18 ± 0.07	—	—
Gaussian noise	91.90	69.02	24.97	83.26	46.58	17.41
Gaussian blur	39.19	18.43	17.48	36.08	18.78	15.18
Motion blur	81.80	59.25	39.19	71.14	52.77	39.00
JPEG	76.94	67.24	39.82	71.95	64.63	43.59
Contrast	94.77	93.53	88.69	88.46	85.64	73.45

Darkening	94.76	93.72	89.83	88.67	86.23	75.56

Table 2. Baseline accuracy (%) per degradation type and severity level for both architectures.

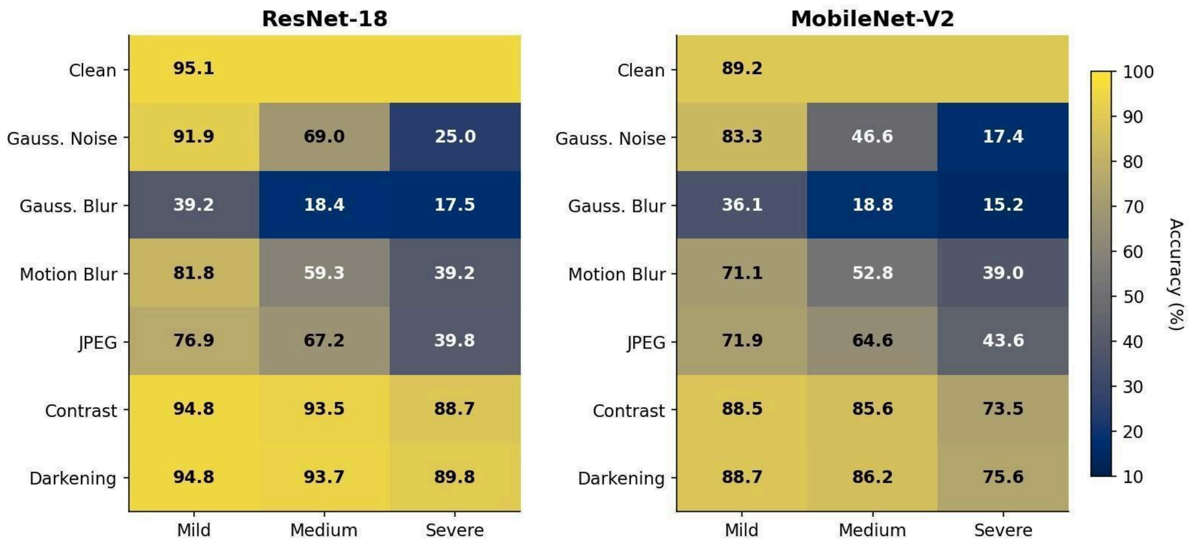


Figure 1. The heatmap of baseline accuracy of ResNet-18 and MobileNet-V2 across the 18-condition degradation grid. The largest collapses of accuracy on both architectures are caused by severity 3 blur and noise.

4.2 Mixed Augmentation

Mixed augmentation is the strategy that stands out from all the rest. For ResNet-18, clean accuracy dropped by 3.00 pp (from 95.09% to 92.09%), while mean robustness improved by 23.71 pp (from 65.59% to 89.30%). For MobileNet-V2, clean accuracy dropped by 3.28 pp (89.18% to 85.90%), while mean robustness improved by 22.56 pp (58.80% to 81.36%). This works out to about 7.9 pp of robustness gained per pp of clean accuracy lost on ResNet-18, which is a tradeoff of approximately 3 pp of clean accuracy for 23 pp of robustness.

On ResNet-18 the gains are statistically significant in all types of degradation except contrast and darkening, where the performance is already close to ceiling/Max and little can be done to improve it. On MobileNet-V2 the improvements are statistically significant on every condition tested including contrast and darkening. The smaller architecture seems to benefit more from the regularisation effect of varied augmentation than larger ones do which is consistent with the broader pattern that compact models gain more from input level diversity than larger ones do.

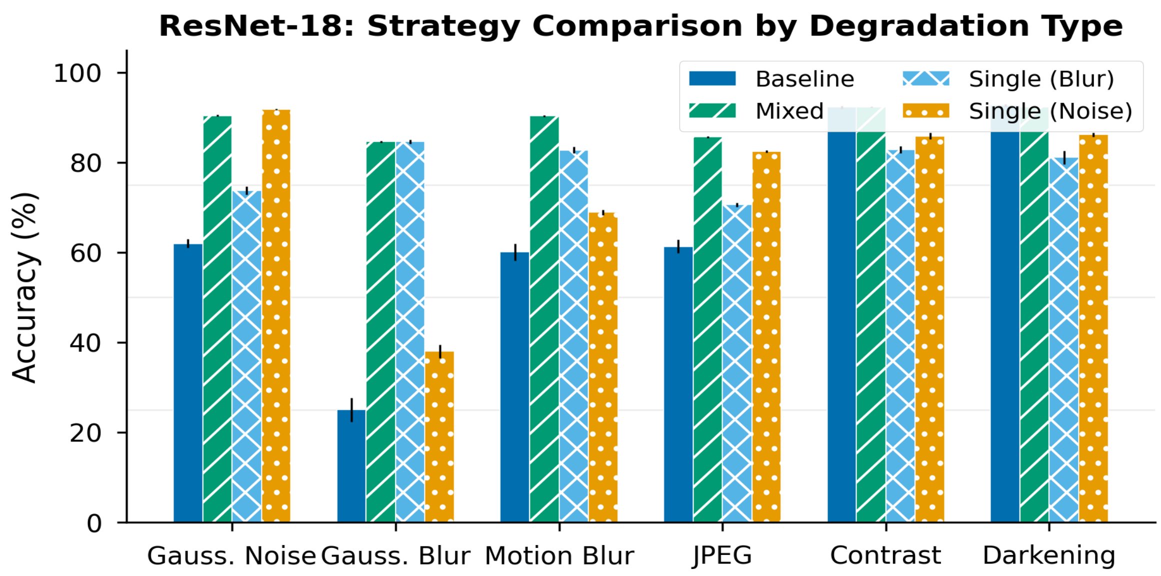


Figure 2. Comparison of the ResNet-18 strategies by type of degradation. Mixed augmentation not only matches or exceeds single-type strategies on most degradations, but also keeps clean-accuracy cost low.

4.3 Single-Type Augmentation

Single-type training improves robustness on its target degradation, but at a higher cost than mixed augmentation.

4.3.1 Training on Gaussian Blur

For Gaussian blur, training on blur directly addresses the blur weakness. ResNet-18 jumps from 25.03% to 84.64% on Gaussian blur ($p < 10^{-7}$, Cohen's d = 38.4), which basically match what mixed augmentation can reach. Motion blur also benefits, improving from 60.08% to 82.79% (+22.71 pp). The cost appears across the remaining conditions. Clean accuracy drops by 9.05 pp for ResNet-18 and 11.20 pp for MobileNet-V2 (both $p < 10^{-5}$). Contrast loses 9.40 pp and darkening loses 11.64 pp on ResNet-18. This is a substantial negative transfer. The model has been pushed to rely on low-frequency features, which helps under blur but harms discrimination under lighting shifts.

MobileNet-V2 shows the same pattern in sharper form: clean accuracy down 11.20 pp ($p < 10^{-6}$), blur up 53.64 pp, contrast down 8.18 pp, darkening down 9.02 pp. The smaller architecture has less capacity to absorb the representational shift, so the interference from blur training is correspondingly larger.

4.3.2 Training on Gaussian Noise

Training on noise is considerably less costly. ResNet-18 reaches 91.83% mean noise accuracy (up from 61.96%, $p < 10^{-7}$, Cohen's d = 42.6) and gives up only 2.02 pp of clean accuracy, approximately one third of the cost of blur training. Noise leaves the spatial structure of the image intact: edges and shapes are still visible, only pixel-level static has been added. Blur removes that structure. The model can adapt to noise without rewriting its feature hierarchy, but it cannot do the same for blur.

The noise-trained model also improves on degradation types which were not used during training. JPEG accuracy increases by 21.15 pp ($p < 10^{-5}$), consistent with the fact that low-quality JPEG artefacts share statistical properties with additive noise around block boundaries. Motion blur improves by 8.83 pp ($p < 0.001$). Noise training therefore acts as a partial regulariser against high-frequency perturbations. Gaussian blur, however, improves by only 12.97 pp, which confirms that noise and blur pose different problems.

MobileNet-V2 follows the same pattern: noise robustness up 36.04 pp, clean accuracy down only 2.18 pp, JPEG transfer of +14.93 pp, contrast loss of 4.97 pp. All effects are smaller in absolute terms, consistent with the smaller model.

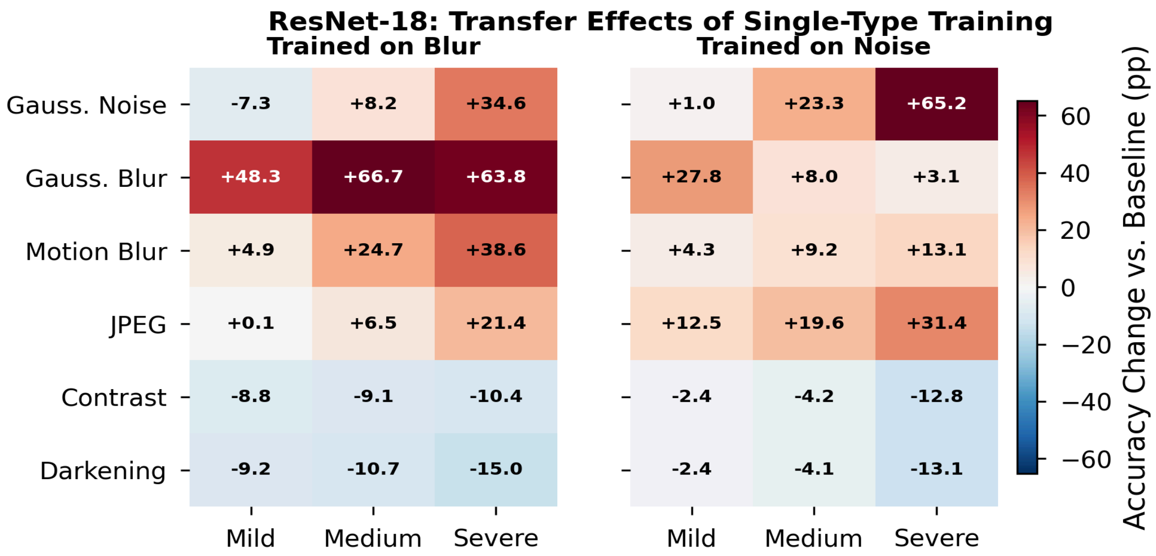


Figure 3. Transfer effects of single-type training (ResNet-18). Left panel: training on blur. Right panel: training on noise. Accuracy change versus baseline in percentage points; positive values (red) indicate improvement, negative values (blue) indicate degradation.

4.4 Severity-Curriculum Augmentation

The curriculum strategy produced the most striking result of the study: a clear split between partial success and catastrophic failure, depending on which degradation type is used.

4.4.1 Curriculum on Gaussian Blur: Catastrophic Collapse

both ResNet-18 and MobileNet-V2 collapse to near-random performance. ResNet-18 clean accuracy drops from 95.09% to 12.77% ± 2.01%, and MobileNet-V2 drops from 89.18% to 11.86% ± 1.59%. This is not measurement error: the models go from state-of-the-art CIFAR-10 performance to barely above the 10% chance baseline for a ten-class problem.

The trained model performs well on only one input distribution: high-severity blurred images. Accuracy increases with blur severity, the inverse of every other strategy (Table 3), while clean images are no longer classified above chance.

Severity	ResNet-18	MobileNet-V2
1	14.30% ± 1.91%	14.37% ± 2.77%
2	42.30% ± 2.42%	45.22% ± 5.54%
3	80.73% ± 0.20%	73.12% ± 0.37%

Table 3. Curriculum-blur trained models tested on Gaussian blur, by severity. The more severe the input the more accurate the model becomes, the opposite of the situation with all other conditions.

Mean robustness is 18.98% ± 1.53% for ResNet-18 and 17.68% ± 1.91% for MobileNet-V2, both well below their respective baselines (65.59% and 58.80%). The augmentation is actively harming the model. The collapse is consistent across all 5 seeds on both architectures, so it is not an outlier or an error. Section 5.2 covers the mechanism in detail.

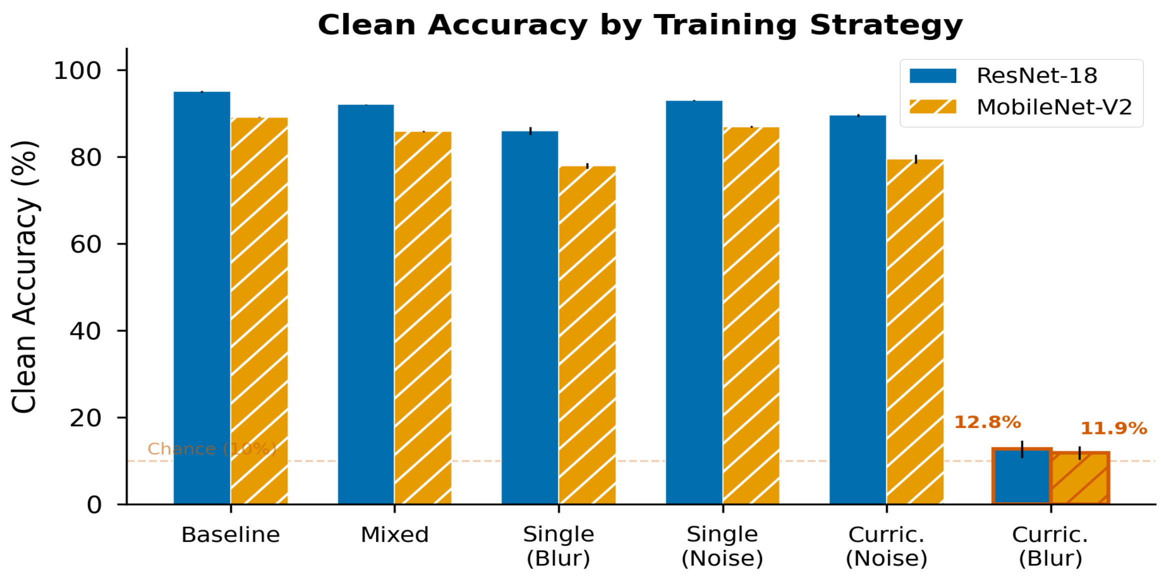


Figure 4. Clean accuracy by training strategy. The curriculum-blur configuration on the right collapses to approximately the chance baseline (dashed orange line) for both architectures.

4.4.2 Curriculum on Gaussian Noise: Marginal

ResNet-18 with curriculum/noise reaches 89.61% clean accuracy (–5.48 pp from baseline) and 90.04% accuracy on sev3 Gaussian noise, which is comparable to single-type/noise. Mean degraded accuracy is 66.90% ± 0.57%. The robustness gain over baseline is only +1.31 pp.

Compared with mixed augmentation (mean degraded 89.30%), curriculum/noise is far behind. Compared with single-type/noise (mean degraded 75.56% at a clean cost of only 2.02 pp), curriculum/noise cost more and perform worse. Curriculum training adds cost without adding benefit, even where it does not fail outright. The ordered exposure is not doing what the Bengio (2009) curriculum hypothesis predicted, at least not at this scale.

there is also a less obvious finding here: curriculum/noise produce large drops on contrast (–24.97 pp on ResNet-18) and darkening (–24.04 pp), far worse than the –6 to –7 pp drops from single-type/noise. The severity ramp, which exposes the model to increasingly intense noise for 100 epochs, causes more severe representational specialisation than random severity sampling does. The specialisation overshoots.

MobileNet-V2 is still vulnerable. Clean accuracy dropped by 9.69 pp (to 79.49%), noise robustness improved by 32.52 pp, but mean degraded accuracy is only 55.97% ± 1.52%, which is below the MobileNet-V2 baseline (58.80%). The contrast and darkening drops are even worse (–30.55 pp and –26.40 pp). The compact architecture is more sensitive to the curriculum progressive specialisation.

Why does curriculum/noise not collapse the way curriculum/blur does? Because noise preserves spatial structure even at maximum severity, while blur destroys it. The mechanism is unpacked in Section 5.2.

4.5 Summary

Table 4 summarises the clean and mean-robustness outcomes across all strategies and architectures.

Strategy	Clean (R18)	Mean rob. (R18)	Clean (MNv2)	Mean rob. (MNv2)
Baseline	95.09% ± 0.17%	65.59% ± 0.78%	89.18% ± 0.07%	58.80% ± 1.19%
Mixed	92.09% (–3.00)	89.30% (+23.71)	85.90% (–3.28)	81.36% (+22.56)
Single / blur	86.04% (–9.05)	79.32% (+13.73)	77.98% (–11.20)	70.44% (+11.64)
Single / noise	93.07% (–2.02)	75.56% (+9.97)	87.00% (–2.18)	69.16% (+10.36)
Curriculum / noise	89.61% (–5.48)	66.90% (+1.31)	79.49% (–9.69)	55.97% (–2.83)
Curriculum / blur	12.77% (–82.32)	18.98% (–46.61)	11.86% (–77.32)	17.68% (–41.12)

Table 4. Summary of clean and mean robustness across all strategies and architectures. Values in parentheses are deltas relative to baseline.

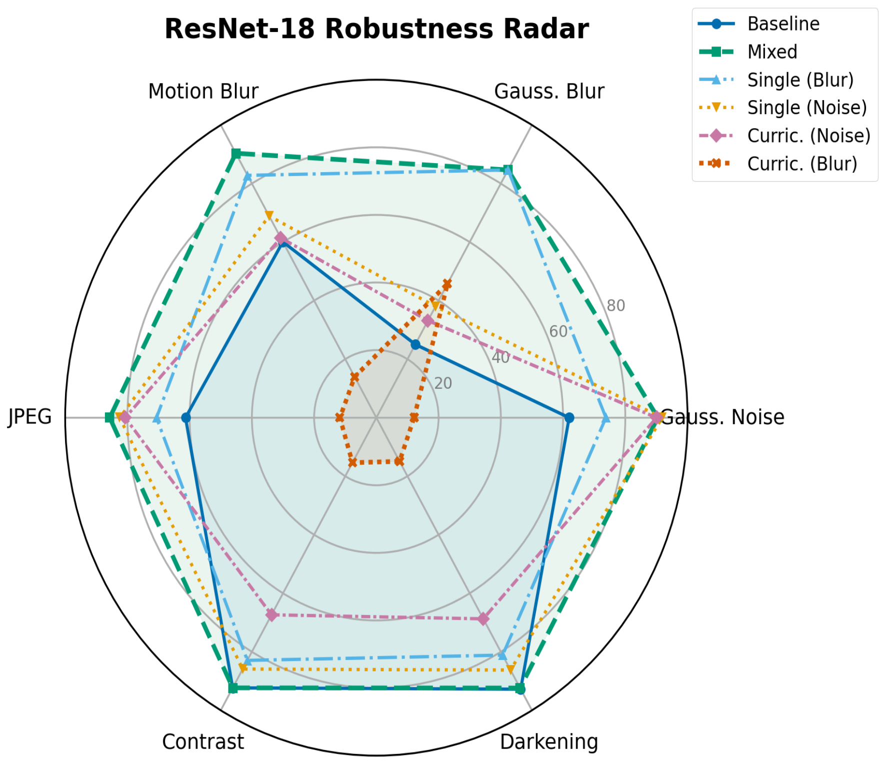


Figure 5a. ResNet-18 robustness radar across the six degradation types. Mixed augmentation produces the largest envelope; curriculum-blur collapses inward across all axes.

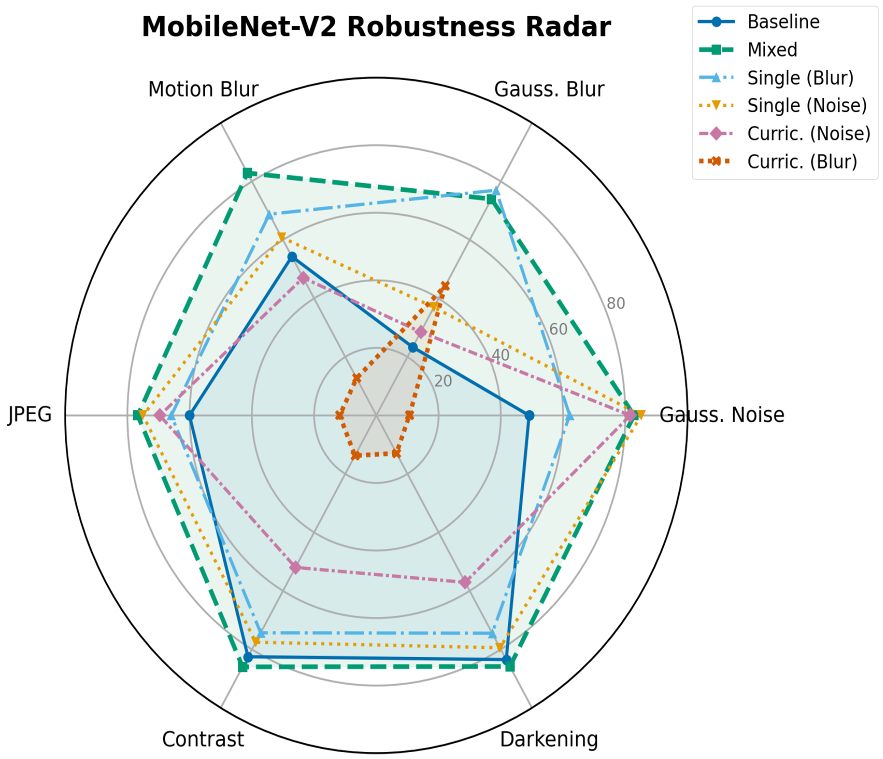


Figure 5b. MobileNet-V2 robustness radar across the six degradation types. The same qualitative pattern as ResNet-18 holds: mixed augmentation dominates, curriculum-blur collapses.

5. Discussion

5.1 Hypothesis Verdicts

Framing note before the verdicts. The proposal anticipated that some hypotheses might not be supported and stated that in that case the analysis would centre on augmentation interference and negative transfer. That is what happened: H2 and H3 were not pre-registered. Each verdict below reports the hypothesis against its original H0/HA formulation from Section 1.4, and the unanticipated findings are treated separately in Sections 5.2 and 5.5.

H1 was supported. With clean-accuracy cost of approximately 3 pp, mixed augmentation hit the highest mean robustness on both architectures: 89.30% ± 0.06% on ResNet-18, and 81.36% ± 0.10% on MobileNet-V2. The gains were statistically significant on all the types of degradation except contrast and darkening in the case of ResNet-18, as baseline performance had already reached the ceiling so there was nothing left to gain. On MobileNet-V2 the gains were significant on all conditions tested, including contrast and darkening.

H2 was not supported: the null hypothesis could not be rejected. The paired comparisons were not significant in the expected direction. On the contrary, under the conditions tested curriculum training performed worse than single-type training. In the case of Gaussian noise, curriculum and single-type training had similar noise robustness, but the curriculum incurred a significantly higher clean-accuracy cost (–5.48 pp vs –2.02 pp on ResNet-18) and a much worse interference on contrast (–24.97 pp vs –6.4 pp on ResNet-18). In the case of Gaussian blur the outcome was sharper still: curriculum training led to a near-total collapse, with clean accuracy falling to 12.77% and mean robustness to 18.98%, both well below baseline. The naive linear-ramp schedule applied to one type of degradation at 32×32 resolution was not able to reproduce the predictions of the curriculum learning hypothesis (Bengio et al., 2009) for image-degradation training in this study. This outcome contradicts the prediction derived from the curriculum learning hypothesis of Bengio et al. (2009) for this setting, and Section 5.2 argues the contradiction is driven by the interaction between the degradation and the input resolution rather than by curriculum ordering as such.

H3 was not supported: the null hypothesis could not be rejected. The ordering prediction, that ResNet-18 would increase more than MobileNet-V2 under mixed augmentation, did not hold consistently across measures. ResNet-18 also achieved slightly more mean robustness when trained on mixed data. but MobileNet-V2 had slightly higher gains under single-type noise training. The two architectures react to the four strategies in a qualitatively similar form, with quantitative differences that do not necessarily run in the same direction. This is taken up in Section 5.5.

H4 was supported. Both single-type strategies showed statistically significant improvement on the trained degradation as well as a statistically significant decrease on at least one untrained degradation at $p < 10^{-4}$. Blur training decreased contrast by 9.4 pp and darkening by 11.6 pp. Noise training decreased both by approximately 6.5 pp. Blur training was 1.5 to 2 times more destructive to lighting performance than noise training, which is consistent with the spatial-structure argument developed in Section 5.2.

5.2 The Curriculum-Blur Failure

The main discovery of this research is the collapse of curriculum/blur. It was not anticipated by the proposal. Both ResNet-18 and MobileNet-V2 drop down to nearly random performance on the clean test set, 12.77% and 11.86%, as compared to a 10% chance baseline of a ten-class problem. The model has not been taught a worse form of classification. It has stopped performing classification at all, except on the condition it was last trained on.

The schedule itself appears benign. The kernel size ramps linearly from 3 to 7 over the first 100 epochs and then holds at 7 for the next 100 epochs. The same shape of schedule applied on Gaussian noise produces a working model. Suboptimal, but working. The difference between the two cases is image size and the spatial nature of the degradation.

Look at what a 7×7 Gaussian blur kernel does to a 32×32 image. On a 32x32 CIFAR image it is destructive. The fine textures vanish. The edges get smeared with their neighbours. Most of the high-frequency content which permits class discrimination at this resolution is simply gone. By epoch 100 the model is being asked to classify images which have lost the features the classification originally relied on.

What happens next is a classic case of catastrophic forgetting (McCloskey & Cohen, 1989; French, 1999). When a neural network is trained on one distribution and then trained exclusively on a second distribution, the weights which encoded the first distribution are getting overwritten by the second. there is no mechanism in standard SGD which protects old representations. The network does whatever minimize the current loss. so if the loss is computed entirely on heavily blurred images for the final 100 epochs, the network reorganises around heavy blur. This is exacerbated by the MultiStepLR schedule. The learning rate does not go to zero, and the long final stage at the highest severity gives the network ample optimisation steps to become specialised.

Curriculum/noise does not fail in the same manner. Noise does not destroy spatial structure even at maximum severity. The high-frequency details and shape information of the model used in clean training are still recoverable through the noise, so the model retains its clean-image features and adds a denoising-like component on top. Blur is different. With blur the structure is removed entirely. There is nothing to preserve. Thus, failure of curriculum is not failure of curriculum learning. It is a failure of curriculum learning combined with a degradation which destroys the input signal at the resolution we are working at. This mechanism is consistent with the texture-bias account of Geirhos et al. (2019). CNNs trained on natural images rely heavily on texture cues, and Gaussian blur removes exactly the high-frequency texture content those cues depend on. A model forced to operate exclusively on heavily blurred 32x32 inputs must therefore abandon the texture-based features which clean training built, which is what the catastrophic forgetting dynamic describes at the weight level. Noise, by contrast, corrupts texture statistics without removing them, so the clean-image features remain useful and the same schedule survives. The collapse observed here can thus be read as a joint prediction of the texture-bias literature and the catastrophic forgetting literature, instantiated at a resolution where the two effects compound.

5.3 Why Mixed Augmentation Works Best

Mixed augmentation is the most effective due to feature diversity. Random sampling on the grid of 18 conditions implies that each minibatch is a heterogeneous blend of degradations and severities. This forces the network to learn features which work across all types of perturbations at once rather than features which are specific to any one particular perturbation. Single-type augmentation has the opposite effect. It pushes the model towards a single perturbation at the expense of the rest. This can be observed directly through the negative transfer to contrast and darkening reported in Section 4.3. Mixed augmentation acts like a multi-task regulariser: instead of memorising one noise pattern the model has to find features that hold up everywhere.

The most informative part of this result is the remarkably small clean-accuracy cost of mixed augmentation (~3 pp), compared to that of single-type training (2 to 11 pp). It suggests that varied exposure of degradation does not significantly contradict clean image classification. The model can accommodate both clean and degraded inputs without major representational compromise.

Indirect comparison with AugMix

According to AugMix (Hendrycks et al., 2020), there is approximately 1 pp of clean-accuracy cost on CIFAR variants. Mixed augmentation here costs about 3. The difference is informative. AugMix uses two extra mechanisms not tested here: augmentation chaining (compositions of multiple operations on a single image) and a Jensen–Shannon consistency loss, which drives the model towards similar predictions across augmented views. The 2 pp gap is what those mechanisms probably contribute over and above degradation exposure. So most of the robustness comes from degradation exposure itself, not the regularisation machinery on top. That matters for deployment cases where the consistency loss is impractical such as edge inference. The simpler approach gets you most of the way there at lower complexity. This is an indirect comparison, as AugMix was not re-trained on the same matrix as here. This is flagged again in Section 5.6. The generalisation of noise training observed in Section 4.3 also parallels Lopes et al. (2019), who showed that localised Gaussian noise augmentation improves robustness to several corruptions the model was never trained on, at low clean-accuracy cost. The present results are consistent with theirs and extend the observation to full-image noise augmentation under a matched multi-seed statistical framework.

5.4 Clean–Robustness Trade-off

For ResNet-18, the four strategies sit at very different points on the tradeoff curve. Mixed augmentation gains 23.71 pp of mean robustness for 3.00 pp of clean accuracy, or more precisely the ratio is 7.9 pp gained per pp lost. This is the most efficient point of the curve. Single-type/noise gained 9.97 pp for 2.02 pp, a ratio of approximately 4.9 pp per pp, a fair trade nonetheless. Single-type/blur gained 13.73 pp for 9.05 pp, a ratio of 1.5 pp per pp, which is much more costly.

Both curriculum strategies do not fit into a meaningful tradeoff framing. Curriculum/blur lose both clean accuracy AND robustness simultaneously, so there is no tradeoff to consider; the strategy is dominated. Curriculum/noise gained only 1.31 pp of robustness at a clean cost of 5.48 pp, which makes it strictly worse on both axes than single-type/noise.

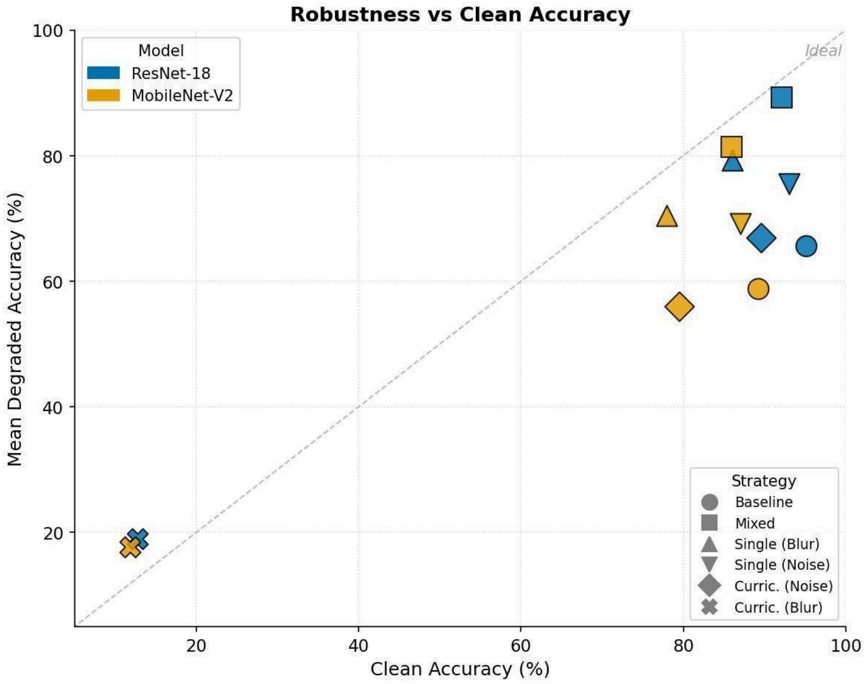


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Figure 6. Robustness vs. clean accuracy. Each marker is one (architecture, strategy) pair. Mixed augmentation (squares) lies closest to the ideal line; curriculum-blur (crosses) collapses to the bottom-left corner.

5.5 Architecture Differences

The response of ResNet-18 and MobileNet-V2 to all four augmentation strategies follows the same qualitative shape. Mixed is best for both. Single-type training operates on its target degradation at a clean-accuracy cost on both. Both architectures collapse under curriculum/blur. Both perform poorly with curriculum/noise. The ranking remains constant across architectures.

The variations are quantitative, and they go in the direction you would expect from capacity. MobileNet-V2 pay more for blur augmentation (–11.20 pp clean accuracy vs –9.05 pp on ResNet-18) and has larger curriculum/noise drops: –9.69 pp vs –5.48 pp on clean accuracy, and a mean-robustness loss of –2.83 pp vs a small gain on ResNet-18. The smaller architecture is less able to absorb the representational changes that accompany the specialising strategies.

Mixed augmentation is the exception: The clean-accuracy cost (–3.00 pp vs –3.28 pp) and the mean-robustness gain (+23.71 pp vs +22.56 pp) are nearly identical across both architectures. Broad exposure produces broad robustness without favouring one architecture over the other, which makes it the most architecture-agnostic strategy of the four.

In summary, the strategy ranking is architecture independent, while MobileNet-V2 pays a consistently higher cost for the specialising strategies.

5.6 Limitations

There are several limitations to flags.

Image scale first. CIFAR-10 is 32×32, and at such a size the more severe grid conditions become extremely destructive. In this case, a 7×7 Gaussian blur kernel covers 22% of the image width, which means the image content is largely destroyed rather than blurred. On a 224×224 ImageNet image the same kernel is much milder. The sev3 conditions mentioned here are not to be interpreted as equivalent to sev3 conditions on full-resolution sensors. Anyway, CIFAR-10 was used because it can provide the statistical framework with the multiple-seed matrix it requires, within the available compute resources. ImageNet-C scale could not be done at this level of resources.

This is a reasonable sample of the failure modes deployed imaging systems encounter, but it omits rain, fog, defocus, lens flare, rolling shutter and others which appear in practice. The set also bounds the transfer analysis; extending the grid is a natural direction for future work.

The samples in mixed augmentation are evenly distributed across the grid of 18 conditions. The weighted scheme, in which milder severities would be over-sampled compared to harsher ones, was not tested. The H1 finding would require qualification if a weighted scheme yielded different mean robustness. This is noted as an open question.

There was no held-out validation split. The 200-epoch training schedule with default hyperparameters, no early stopping, final checkpoint reported. This was a deliberate choice, to keep the test set truly held out over the entire matrix, but it leaves the training pipeline with no in-training signal, either to over- or under-fit. There was also no use of label smoothing. The optimiser setup is standard for these architectures on CIFAR-10.

A more general assertion about curriculum learning as a class of methods would require a broader sweep of schedules, which was beyond the scope of this study.

The AugMix comparison in Section 5.3 is indirect. AugMix was not re-trained with the matrix used here. The comparison is based on the clean-accuracy values from the original paper, and therefore should be interpreted as indicative rather than as a controlled benchmark.

5.7 What This Study Adds

Relative to the existing literature, the findings are sorted into three categories. Consistent with previous work. The baseline gap between clean and degraded accuracy, largest for blur and noise, reproduces the central observation of Hendrycks and Dietterich (2019) and the quality-reduction results of Dodge and Karam (2016) at CIFAR scale. The need for multi-seed statistical validation confirmed itself in practice: seed-level variance was small but non-zero, exactly the regime Bouthillier et al. (2021) warn about, and the earlier undertraining artefact described in Section 1.1 shows how easily single-run comparisons mislead. Contradicting previous work. The curriculum results contradict the prediction derived from Bengio et al. (2009) for this setting. Ordered easy-to-hard exposure did not outperform random exposure on the same degradation, and for Gaussian blur it failed catastrophically. The contradiction is scoped: Section 5.2 argues it arises from the interaction between a structure-destroying degradation and a 32x32 input resolution, not from curriculum ordering in general. Within that scope, however, the result is a clear counterexample to the assumption that a curriculum is a safe default. Extending previous work. Three extensions are offered. First, the comparison isolates degradation exposure from the additional machinery of methods like AugMix (Hendrycks et al., 2020), and shows that exposure alone delivers most of the robustness benefit at CIFAR scale. Second, the cost side of augmentation, usually underreported, is quantified for every strategy, which makes the strategies directly comparable on a clean-robustness tradeoff. Third, the transfer analysis extends the noise-augmentation observations of Lopes et al. (2019) with a documented negative-transfer pattern: single-type training reliably harms lighting conditions it was never trained on. For practitioners the implication is that augmentation strategies should be selected on both axes of the tradeoff, and that mixed exposure is the safest default at this scale.

6. Conclusion

This capstone evaluated three training-time data augmentation methods for CNN robustness on CIFAR-10. There were two architectures, five seeds, eighteen evaluation conditions and paired statistical comparisons. The aim was not only to determine whether each strategy enhances robustness but also the cost of each strategy, and whether that cost is justified.

The main conclusion is that mixed augmentation dominates. Sampling type and severity of degradation uniformly at random across the entire evaluation grid resulted in approximately 23 pp of average robustness gain for approximately 3 pp of clean-accuracy loss. Mixed augmentation is by far the strategy most likely to succeed on the tradeoff curve compared to the other strategies tested. The clean-accuracy cost is minimal enough that it does not significantly conflict with clean image classification, and the robustness gain extends to all types of degradation included in the evaluation grid.

Single-type augmentation works on its target degradation but the cost is paid in other areas. Both blur-trained and noise-trained models show statistically significant negative transfer to contrast and darkening. The blur-trained model incurs the highest clean-accuracy cost, approximately 9 pp on ResNet-18 and 11 on MobileNet-V2. The noise-trained model is less extensive, approximately 2 pp and generalises to other degradations such as JPEG compression and motion blur, which it had never been trained on. This generalisation pattern was the cleanest secondary finding of the study.

The most striking outcome is the curriculum/blur breakdown. Both architectures are dropped from state-of-the-art clean performance to near-random performance by a linear severity ramp on Gaussian blur applied to 32×32 images. The ResNet-18 stopping point is 12.77% clean accuracy. MobileNet-V2 ends at 11.86%. Both are only slightly above the 10% chance threshold of a ten-class problem. Section 5.2 examined the mechanism in detail. The brief version is that at this resolution a 7×7 kernel covers 22% of the image, late-curriculum updates effectively erase the clean-image features through catastrophic forgetting, and the model is left with nothing to fall back on. The same schedule applied to Gaussian noise does not produce this collapse because noise does not destroy the spatial structure the clean-image features were built on. The curriculum failure here is not a failure of curriculum learning as a class of methods. It is a failure of curriculum learning combined with a degradation which destroys the input signal at the resolution being used.

The practical implication is quite clean. Mixed augmentation should be the default for CIFAR-scale images using these architectures. Wide robustness, low clean-accuracy cost, behaviour nearly identical on both architectures. Single-type training is worth considering in cases where the deployment distribution is narrow and known a priori. Naive linear-ramp forms of curriculum scheduling should likely simply be avoided in particular in cases where the degradation destroys the spatial structure of the input.

These results suggest several directions for future work. Curriculum schedules other than the linear ramp (cosine, stepped, validation-gated) were not tested and may behave differently. Mixed augmentation with weighted sampling (milder severities oversampled compared to harsher ones) was also not tested. Replication at higher resolution would clarify whether the curriculum/blur collapse is essentially about the schedule or essentially about the resolution-degradation interaction. A direct comparison could replace the indirect comparison with AugMix in Section 5.3 by re-training AugMix under the same matrix used here.

The gap between clean and degraded performance of lightweight CNNs is real, large and unresolved. This research does not close that gap entirely, but it identifies a strategy which closes most of it at low cost, and a strategy which does the reverse of what it was designed to do.

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